Data Warehousing & Mining: Unit-5

Basics of Data Mining

5.1 Data Mining

Data mining is the process of extracting meaningful hidden patterns and knowledge from large datasets using machine learning and statistical techniques to support informed decision-making.

Data mining involves followings:

-data preparation

-selecting relevant data

-building models, and

-evaluating results to uncover trends, relationships, and anomalies that are too complex for traditional analysis.

5.2 Data Mining Key Techniques

1. Classification:

Assigning data instances to predefined categories, such as identifying spam emails.

2. Clustering:

Grouping similar data points together, like segmenting customers based on purchasing habits.

3. Regression:

Predicting a continuous numerical value, such as forecasting future sales based on historical data.

4. Association Rule Mining:

Discovering relationships between variables, like iden tifying products frequently bought together in a market basket analysis.

5. Anomaly Detection:

Identifying unusual data points that deviate from the norm, such as detecting fraudulent transactions.

5.3 Applications of Data Mining

1. Marketing:

Understanding customer preferences and targeting marketing campaigns more effectively.

2. Healthcare:

Identifying factors contributing to diseases or predicting patient outcomes.

3. Finance:

Detecting fraudulent activities, assessing credit risk, and managing risk.

4. Retail:

Analyzing sales data to optimize inventory, understand customer buying patterns, and make informed decisions on product placement.

5.4 Data Mainlining Algorithms

Data mining algorithms are computational methods used to extract meaningful patterns and insights from large datasets. They build models from data to identify trends, which helps in making predictions, classifying information, or grouping similar items. Examples include the k-means algorithm for clustering, the Naive Bayes algorithm for classification, and the Apriori algorithm for association rule mining.

Types of data mining algorithms

5.4.1 Classification:

Predicts a target variable by assigning data points to predefined categories.

- C4.5 Algorithm

: A decision tree-based classifier.

- Naive Bayes : A probabilistic classifier that assumes feature

independence.

- Support Vector Machines (SVM): A supervised learning model that classifies data into two groups.

- k-Nearest Neighbors (KNN): A simple algorithm that classifies data

based on its nearest neighbors in the feature space.

5.4.2 Clustering:

Groups similar data points together without prior knowledge of the groups (unsupervised learning).

- k-means

: Partitions data into a specified number of clusters based on their

distance from cluster centers.

- Expectation-Maximization (EM): A statistical algorithm for finding

maximum likelihood or maximum a posteriori estimates of parameters.

5.4.3 Association Rule:

Discovers relationships between items in a dataset, often used in "market basket" analysis.

- Apriori Algorithm : Finds frequent itemsets in transactional databases and

generates association rules.

5.4.4 Prediction and Regression:

Predicts a continuous value.

- CART (Classification and Regression Trees): Creates a tree structure for

both classification and regression tasks.

5.4.4 Ensemble Methods:

Combines multiple machine learning models to improve overall performance.

- AdaBoost: An algorithm that builds a strong classifier by combining many weak

classifiers.

5.5 Explanation of Some Important Data Mining Algorithms

Here, few Data Mining Algorithms are discussed below.

1. Statistical Procedure Based Approach

There are two main phases present to work on classification. That can easily identify the statistical community.

Generally, statistical procedures have to characterize by having a precise fundamental probability model. That used to provides a probability of being in each class instead of just a classification. Also, we can assume that techniques will use by statisticians. Hence some human involvement has to assume with regard to variable selection. Also, transformation and overall structuring of the problem.

2. Machine Learning-Based Approach

It covers automatic computing procedures based on logical or binary operations. It learn a task from a series of examples. Here, we focus on decision -tree approaches. As classification results come from a sequence of logical steps. These classification results are capable of representing the most complex problem given. Such as genetic algorithms and inductive logic procedures (I.LP.) are currently under active improvement.

This approach aims to generate classifying expressions. That is simple enough to understand by the human and mimic human reasoning to provide insight into the decision process. Like statistical approaches, background knowledge may use in development. But the operation is assumed without human interference.

3. Neural Network

The field of Neural Networks has arisen from diverse sources. That is ranging from understanding and emulating the human brain to broader issues. That is of copying human abilities such as speech and use in various fields. Such as banking, in classification program to categorize data as intrusive or normal.

Generally, neural networks consist of layers of interconnected nodes. That each node producing a non-linear function of its input and input to a node may come from other nodes or directly from the input data. Also, some nodes are identified with the output of the network.

On the basis of this, there are different applications for neural networks present involve d in recognizing patterns and making simple decisions about them. In airplanes, we can use a neural network as a basic autopilot. In which input units read signals from the various instruments and output units. That modifying the plane’s controls appropriately to keep it safely on course. Inside a factory, we can use a neural network for quality control.

3. Classification Algorithms in Data Mining

It is one of the Data Mining. That is used to analyze a given data set and takes each instance of it. It assigns this instance to a particular class. Such that classification error will be least. It is used to extract models. That define s important data classes within the given data set. Classification is a two-step process.

During the first step, the model is created by applying a classification algorithm. That is on training data set. Then in the second step, the extracted model is tested against a predefined test data set. That is to measure the model trained performance and accuracy. So classification is the process to assign class label from a data set whose class label is unknown.

4. ID3 Algorithm

This Data Mining Algorithms starts with the original set as the root hub. On every cycle, it emphasizes through every unused attribute of the set and figures. That the entropy of attribute. At that point chooses the attribute. That has the smallest entropy value.

The set is S then split by the selected attribute to produce subsets of the information. This Data Mining algorithms proceed to recurse on each item in a subset. Also, considering only items never selected before. Recursion on a subset may bring to a halt in one of these cases:

- Every element in the subset belongs to the same class (+ or -), then the node is turned into a leaf and
- labeled with the class of the examples
- If there are no more attributes to s elect but the examples still do not belong to the same class. Then the node is turned into a leaf and labeled with the most common class of the examples in that subset.
- If there are no examples in the subset, then this happens. Whenever parent set found to be matching a specific value of the selected attribute.
- For example, if there was no example matching with marks >=100. Then a leaf is created and is labeled with the most common class of the examples in the parent set.

Working steps of Data Mining Algorithms is as follows,

- Calculate the entropy for each attribute using the data set S.
- Split the set S into subsets using the attribute for which entropy is minimum.
- Construct a decision tree node containing that attribute in a dataset.
- Recurse on each member of subsets using remaining attributes.

5. C4.5 Algorithm

C4.5 is one of the most important Data Mining algorithms, used to produce a decision tree which is an expansion of prior ID3 calculation. It enhances the ID3 algorithm. That is by managing both continuous and discrete properties, missing values. The decision trees created by C4.5 is used for grouping and often referred to as a statistical classifier.C4.5 creates decision trees from a set of training data same way as an Id3 algorithm. As it is a supervis ed learning algorithm it requires a set of training examples. That can see as a pair: input object and the desired output value (class).

This algorithm analyzes the training set and builds a classifier. That must have the capacity to accurately arrange both training and test cases. A test example is an input object and the algorithm must predict an output value. Consider the sample training data set S=S1, S2,…Sn which is already classified. Each sample Si consists of feature vector (x1,i, x2,i, …, xn,i). Where xj represent attributes or features of the sample. The class in which Si falls. At each node of the tree, C4.5 selects one attribute of the data. That most efficiently splits its set of samples into subsets such that it results in one class or the other.

The splitting condition is the normalized information gain. That is a non- symmetric measure of the difference. The attribute with the highest information gain is chosen to make the decision.

General working steps of algorithm is as follows,

Assume all the samples in the list belong to the same class. If it is true, it simply creates a leaf node for the decision tree so that particular class will select . None of the features provide any information gain. If it is true, C4.5 creates a decision node higher up the tree using the expected value of the class. An instance of previously- unseen class encountered. Then, C4.5 creates a decision node higher up the tree using the expected value.

6. K Nearest Neighbors Algorithm

The closest neighbor rule distinguishes the classification of an unknown data point. That is on the basis of its closest neighbor whose class is already known. The nearest neighbor is computed on the basis of estimation of k. That indicates how many nearest neighbors are to c onsider to characterize. It makes use of the more than one closest neighbor to determine the class. In which the given data point belongs to and so it is called as KNN. These data samples are needed to be in the memory at the runtime.

The training points are assigned weights. According to their distances from sample data point. But at the same, computational complexity and memory requirements remain the primary concern. To overcome memory limitation size of data set is reduced . For this, the repeated patterns. That don't include additional data are also eliminated from training data set.

To further enhance the information focuses which don't influence the result. That are additionally eliminated from training data set.

The NN training data set can organize utilizing different systems. That is to enhance over memory limit of KNN. The KNN implementation can do using ball tree, k-d tree, and orthogonal search tree. The tree-structured training data is further divided into nodes and techniques. Such as NFL and tunable metric divide the training data set according to planes. Using these algorithms we can expand the speed of basic KNN algorithm. Consider that an object is sampled with a set of different attributes.

Assuming its group can determine from its attributes. Also, different algorithms can use to automate the classification process. In pseudo code, k-nearest neighbor algorithm can express,

K ← number of nearest neighbors

For each object Xin the test set do

calculate the distance D(X,Y) between X and every object Y in the training set

neighborhood ← the k neighbors in the training set closest to X

X.class ← SelectClass (neighborhood)

7. Naïve Bayes Algorithm

The Naive Bayes Classifier technique is based on the Bayesian theorem. It is partic ularly used when the dimensionality of the inputs is high. The Bayesian Classifier is capable of calculating the possible output. That is based on the input. It is also possible to add new raw data at runtime and have a better probabilistic classifier. This classifier considers the presence of a particular feature of a class. That is unrelated to the presence of any other feature when the class variable is given.

For example, a fruit may consider to be an apple if it is red , round. Even if these features depend on each other features of a class. A naive Bayes classifier considers all these properties to contribute to the probability. That it shows this fruit is an apple.

Algorithm works as follows,

Bayes theorem provides a way of calculating the posterior probability, P(c|x), from P(c), P(x), and P(x|c). Naive Bayes classifier considers the effect of the value of a predictor (x) on a given class (c). That is independent of the values of other predictors.

P(c|x) is the posterior probability of class (target) given predictor (attribute) of class.

P(c) is called the prior probability of class.

P(x|c) is the likelihood which is the probability of predictor of given class.

P(x) is the prior probability of predictor of class.

5.5 Steps involved in Data Mining (Process)

The data mining process typically follows these steps:

1. Set Objectives:

Define the business goals and questions the data mining process should answer. It defines the business goals and key objectives so the da ta mining effort directly address business problem.

2. Data Selection:

Identify and select the specific data sets relevant to the defined objectives. Or choose the relevant data for the analysis from the larger datasets.

3. Data Understanding:

Collect and gain an initial understanding of the data, exploring its properties and identifying potential quality issues or areas for improvement.

4. Data Preparation:

Clean the raw data by handling missing values, removing duplicates, and transforming it into a usable format for analysis. This extensive step involves several sub-processes:

Data Cleaning: Remove errors, fill in missing values, and address outliers.

Data Integration: Combine data from multiple sources into a single dataset.

Data Transformation: Convert data into suitable formats for the modeling stage, which may include scaling or encoding variables.

5. Model Building and Pattern Mining:

Apply algorithms to find patterns, trends, and relationships within the data. It applies appropriate data mining techniques, such as algorithms for classification, regressi on, or clustering, to build models that reveal patterns and relationships.

6. Evaluation and Implementation or Deployment:

Evaluate the patterns and insights discovered and present them, often using data visualizations, for interpretation and implementation.

7. Maintenance:

Continuously monitor and refine the deployed models and processes to ensure their effectiveness over time.

5.5 Knowledge Discovery in Databases (KDD) Process

The Knowledge Discovery in Databases (KDD) process is a multi -step approach to extract valuable, previously unknown insights from large datasets. It is a process of discovering useful knowledge and insights from large and complex datasets. The KDD process involves a range of techniques and methodologies, including data preprocessing, data transformation, data mining, pattern evaluation, and knowledge representation.

It involves data selection,

preprocessing (including cleaning and transformation), data mining to find patterns, pattern evaluation to assess their significance, and knowledge presentation to make the results understandable. This iterative process aims to turn raw data into actionable knowledge.

KDD and data mining are closely related processes, with data mining being a key component and subset of the KDD process. Th

is process aims to identify hidden patterns, relationships,

and trends in data that can be used to make predictions, decisions, and recommendations.

It is a broad and interdisciplinary field used in various industries, such as finance, healthcare, marketing, e -commerce, etc. KDD is very important for organizations and businesses as it enables them to derive new insights and knowledge from their data, which can be further used to improve decision- making, enhance the customer experience, improve business processes, support strategic planning, optimize operations, and drive business growth.

5.5.1 KDD Process in Data Mining

The following are the main steps involved in the KDD process -

- Data Selection - The first step in the KDD process is identifying and selecting the relevant data for analysis. This involves choosing the relevant data sources, such as databases, data warehouses, and data streams, and determining which data is required for the analysis.
- Data Preprocessing - After selecting the data, the next step is data preprocessing. This step involves cleaning the data, removing outliers, and removing missing, inconsistent, or irrelevant data. This step is critical, as the data quality can significantly impact the accuracy and effectiveness of the analysis.
- Data Transformation - Once the data is preprocessed, the next step is to transform it into a format that data mining techniques can analyze. This step involves reducing the data dimensionality, aggregating the data, normalizing it, and discretizing it to prepare it for further analysis.
- Data Mining - This is the heart of the KDD process and involves applying various data mining techniques to the transformed data to discover hidden patterns, trends, relationships, and i nsights. A few of the most common data mining techniques include clustering, classification, association rule mining, and anomaly detection.
- Pattern Evaluation - After the data mining, the next step is to evaluate the discovered patterns to determine their usefulness and relevance. This involves assessing the quality of the patterns, evaluating their significance, and selecting the most promising patterns for further analysis.
- Knowledge Representation - This step involves representing the knowledge extracte d from the data in a way humans can easily understand and use. This can be done through visualizations, reports, or other forms of communication that provide meaningful insights into the data.
- Deployment - The final step in the KDD process is to deploy the knowledge and insights

gained from the data mining process to practical applications. This involves integrating the knowledge into decision- making processes or other applications to improve organizational efficiency and effectiveness.

5.6 Introduction to Web Mining

Web mining is the process of extracting and analyzing large amounts of data from the World Wide Web. It refers to the process of discovering and extracting useful information from a large amount of data available on the World Wide Web.
Web mining is the process of using data mining techniques to extract valuable information from web data, such as web pages, links, and server logs. It applies data mining to the web to find patterns, trends, and insights about users and industries by collecting and analyzing data from various sources, including content, structure, and usage data.

5.6.1 Types of web mining

- Web Content Mining:

Focuses on extracting useful information from the content of web pages, such as text, images, and multimedia. It uses techniques like text mining and natural language processing to analyze content for things like sentiment analysis and product recommendations.

- Web Structure Mining:

Analyzes the link structure of the web to identify relationships between web pages. It helps to understand the way pages are connected, which can aid in search engine optimization and indexing.

- Web Usage Mining

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Extracts information from data generated by user interactions with web servers, such as server access logs, client -side cookies, and user profiles. The goal is to understand user behavior and predict how they interact with the web.

5.6.2 The Web Mining Process

- Resource Identification:

Identify the necessary resources, like server logs or web pages, for information extraction.

- Pre-processing:

Clean and prepare the raw data by selecting relevant information from the sources and transforming it into a usable format.

- Pattern Discovery:

Use data mining and machine learning techniques to find patterns and discover knowledge within the pre-processed data.

- Analysis:

Interpret and validate the discovered patterns to turn them into actionable insights.

5.6.3 Application of web Mining

Web mining has numerous applications in various fields, including business, marketing, e-commerce, education, healthcare, and more. Some common applications of web mining include –

- Marketing and Advertising–

Web mining is used to analyze consumer behavior, identify trends, and personalize marketing campaigns. This includes targeted advertising, product recommendation, and customer segmentation.

- Business Intelligence-

Web mining is used to extract valuable insights from web data, including competitor analysis, market trends, and customer preferences.

- E-commerce–

Web mining is used to analyze user behavior on e -commerce websites, including purchase history, search queries, and clickstream data. This information can be used to optimize website design, personalize product recommendations, and improve customer experience.

- Fraud Detection-

Web mining is used to detect fraudulent activities, such as credit card fraud, identity theft, and online scams. This includes analyzing user behavior patterns, detecti ng anomalies, and identifying potential security threats.

- Social Network Analysis-

Web mining is used to analyze social media data and identify social networks, communities, and influencers. This information can be used to understand social dynamics, sentiment analysis, and targeted advertising.

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